

BALA SUBRAMANIAN L

Mechanical Design Engineer

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PROFESSIONAL SUMMARY

Results-driven Mechanical Design Engineer with 2+ years of R&D experience in engineering design for pump packages, hydraulic systems, and related assemblies using SolidWorks. Skilled in Reverse Engineering, Benchmark Study, GD&T, Tolerance stack-up analysis, DFMEA, DFM/DFA, BOM management, PLM tools (ENOVIA/SAP), New Product Development and technical documentation that supports product safety, release engineering, and consistent design quality. Delivers reliable mechanical assemblies for waste water pump applications through prioritization, and continuous improvement.

TECHNICAL SKILLS

Engineering Tools: AutoCAD, SolidWorks, MS Office, PLM (ENOVIA), SAP

Core Engineering: Mechanical design, Reverse engineering, Benchmark study & competitor analysis, BOM preparation, Root cause analysis (RCA), Hydraulic design & performance optimization, Failure analysis & reliability improvement, Material selection, New product development (NPD), Cost estimation.

Professional Skills: 3D Modeling, Surface Modelling, Sheet Metal Design, Weldments, Exploded Views, 2D drafting, GD&T, ASME Y14.5, DFMEA, DFM, DFA, Tolerance Stack-up Analysis, Prototype Testing & Validation, Engineering change management (ECM), ANSI/ASME, API, ISO Standards, Design calculations.

WORK EXPERIENCE

Design Engineer — CRI Pumps Private Limited

April 2026 – Present

Coimbatore, Tamil Nadu

- Engineered precision pump components in SolidWorks, improving first-pass accuracy and cutting design iteration cycles by 30%.
- Developed complete mechanical assemblies, surface models, and exploded views for Sewage/Drainage pumps, Submersible Mixer pumps, Jet-type Aerators ensuring structural integrity and operational reliability.
- Conducted benchmark study and competitive analysis of 5+ market-leading pump brands, applying reverse engineering techniques to identify design gaps and implement targeted performance improvement opportunities.
- Executed design verification and drawing reviews in SolidWorks, then modify the models based on findings.
- Collaborated with cross-functional teams (Quality, Manufacturing) to resolve engineering issues, drive product validation, support prototype development, and achieve NPD milestones.

Associate Engineer — CRI Pumps Private Limited

September 2025 – March 2026

Coimbatore, Tamil Nadu

- Developed CAD models and technical drawings in SolidWorks, aligning GAD and CSD outputs with ISO standards.
- Generated and maintained item codes and EBOMs for 500+ components in ENOVIA PLM, ensuring data accuracy and strengthening traceability across the product lifecycle.
- Prepared General Arrangement Drawings (GAD) and Technical Data Sheets (TDS) for marketing, improving customer response and proposal turnaround time.
- Administered change requests (CR) and change orders (CO) through Engineering change Management (ECM) workflows, reducing field issues by 20%.

Graduate Engineer Trainee (GET) — CRI Pumps Private Limited

June 2024 – August 2025

Coimbatore, Tamil Nadu

- Developed manufacturing drawings for machining and casting processes and applying DFM principles.
- Maintained BOMs, test reports, and inspection records to support accurate configuration control and audit readiness.
- Strengthened design fundamentals and design calculations through structured training, hydraulic testing procedures and exposure to R&D workflows across 3 product families.

KEY PROJECTS

Project 1: Optimization of Oil Chamber System to Improve Mechanical Seal Performance in Dry Pit Sewage Pumps

- Designed and implemented an innovative oil chamber system for vertical and horizontal dry pit sewage pumps improving mechanical seal lubrication and achieving a significant reduction in mechanical seal failure rates by 20%.
- Validated design effectiveness across 10 pump models (low to high kW range) under controlled R&D conditions ensuring reliability and operational stability before production release.

Project 2: Upgrading Sewage pump Efficiency class from IE2 to IE3 Class

- Engineered and developed sewage pump models aligned with IE3 energy efficiency standards as part of R&D responsibilities, implementing design modifications that improved motor-pump system performance by 5% more than IE2.
- Optimized existing sewage pump designs to meet IE3 efficiency requirements, contributing to energy savings and regulatory compliance for the product range.

Project 3: Design & Development of Mixer pump Installation Stand for Type 11T & 12P & Auto Coupling Tank Configuration in Wastewater applications

- Designed and developed structural installation stands, applying sheet metal design and weldment modelling techniques, for tank-mounted mixer pump configurations (11T and 12P, auto coupling type), managing end-to-end NPD from concept through final design validation.
- Applied tolerance stack-up analysis and material selection principles to ensure structural integrity, long-term reliability in wastewater environments.

EDUCATION

B.E. – Mechanical Engineering

2021 – 2024

Coimbatore Institute of Technology, Coimbatore, Tamil Nadu | CGPA: 7.8/10

Diploma in Mechanical Engineering

2019 – 2021

S. Thangapazham Polytechnic College, Vasudevanallur, Tamil Nadu | Percentage: 99%

Higher Secondary Education – 12th

2019

H.N.U.C. Hr. Sec. School, Puliyangudi, Tamil Nadu | Percentage: 81.1%

SSLC

2017

H.N.U.C. Hr. Sec. School, Puliyangudi, Tamil Nadu | Percentage: 94.8%

CERTIFICATIONS & TRAINING

- SolidWorks Design Associate (CSWA) – DASSAULT SYSTEMES
 - AutoCAD 2D/3D Certification (level 1 & 2) – Autodesk
 - Inspection and Quality Control in Manufacturing - The Engineers Edge Institute of NDT
 - Rapid Manufacturing – NPTEL
 - Work System Design - NPTEL
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ADDITIONAL INFORMATION

Languages: Tamil (Native) | English (Professional)

Location: Coimbatore, Tamilnadu – Open to Relocation